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Rizal, Iloilo City, Philippines

AUTOMATED TRANSACTION RECORDING AND INVENTORY MANAGEMENT SYSTEM FOR CAREMED MEDICAL SUPPLY ENTERPRISE

(ATRIMS-CareMed)

A Business Process Automation Plan for Streamlining Sales Transactions, Rental Recording, and Inventory Control

A BAM - 285 Final Course Work

Presented to the Faculty of the

College Information Technology Education

Phinma University of Iloilo

Rizal, Iloilo City

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Information Technology

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EXECUTIVE SUMMARY

CareMed Medical Supply Enterprise records sales in handwritten logbooks, rentals on printed forms and Facebook Messenger threads, and inventory in an Excel worksheet that is updated only at the end of the day, with oxygen tanks physically counted before every Prime Gas refill. These methods were adequate when the business was smaller, but the growing volume of daily sales, rentals, and refills has exposed their weaknesses. Transactions are recorded late or omitted, recorded stock disagrees with physical counts, a rental item is occasionally promised to two customers, and the owner spends hours reconciling notebooks, worksheets, and chat histories at every cutoff. With no transaction-level audit trail, these discrepancies cannot be traced to a source.

The proposed Automated Transaction Recording and Inventory Management System for CareMed (ATRIMS-CareMed) is designed to eliminate these manual checkpoints through an integrated, web-based platform. It automates point-of-transaction recording of sales and rentals, real-time stock deduction and restoration, low-stock alerts, refillable oxygen tank status tracking (Full / Rented / Empty-For-Refill), chronological audit logs, and one-click exportable reports. Every stock change is logged with a timestamp and user

identifier, and every transaction receives a reference number, giving the owner a complete, traceable history.

Using Stakeholder Analysis, SIPOC modeling, As-Is and To-Be Process Mapping, Fishbone 4P Root-Cause Analysis, Requirements Engineering, and Cost-Benefit Justification, this coursework sets design targets of about 2 minutes or less per transaction (from about 10 minutes) and under 5 minutes for monthly reports (from about 3 hours). All 13 functional test cases passed. These time figures are design targets that still have to be confirmed with measurements from the two-week parallel pilot.

Table of Contents

CHAPTER I INTRODUCTION 1

1.1 Background of the Project 1

1.2 Statement of the Problem 3

1.2.1 General Problem 3

1.2.2 Specific Problems 3

1.3 Objectives of the Project 5

1.3.1 General Objective 5

1.3.2 Specific Objectives 5

1.4 Significance of the Project 6

1.5 Purpose and Description 7

1.6 Scope and Limitations 9

CHAPTER II REVIEW OF RELATED LITERATURE AND SYSTEMS 11

2.1 Related Literature 11

2.1.1 The automation of business processes in small and medium-sized enterprises 11

2.1.2 Digital records and the quality of information .. 11

2.1.3 The management of medical inventory and rentals . 12

2.1.4 Audit Trails and Requirements Analysis 13

2.2 Related Systems 14

2.3 Synthesis 14

CHAPTER III METHODOLOGY, RESULTS, AND DISCUSSION 16

3.1 Requirements Analysis 16

3.1.1 Stakeholder Analysis 16

3.1.2 SIPOC Analysis 17

3.1.3 Current-State Process 18

3.1.4 Problems Identified 19

3.1.5 Root-Cause Analysis (Fishbone 4P Diagnostic) 20

3.1.6 Requirements Engineering 21

3.2 Framework of the Project 24

3.2.1 Proposed Future-State Process 25

3.2.2 Before and After Comparison 26

3.2.3 Proposed System Modules 26

3.2.4 Automation Rules Engine 27

3.2.5 Sample Sales Transaction Record 29

3.2.6 Proposed Dashboard and Visual Layout 29

3.3 Development and Testing 32

3.3.1 Prototype Layouts 32

3.4 Implementation Plan 33

3.4.1 Project Phases 33

3.4.2 Implementation Schedule (Gantt Representation) .. 34

3.4.3 Risk Analysis and Mitigation Matrix 35

3.4.4 Cost-Benefit Analysis and ROI Calculation 36

3.4.5 Change Management and Training Framework 42

CHAPTER IV RESULTS AND DISCUSSION 44

4.1 Implementation 44

4.1.1 Presentation Layer 44

4.1.2 Application Layer 45

4.1.3 Data Layer 45

4.1.4 Deployment Setup 46

4.2 Development Tools 46

4.3 Hardware Requirements 48

4.4 Software Requirements 48

4.5 Inputs and Outputs of the Study 49

4.6 Results Interpretation and Analysis 51

4.6.1 Achievement of Objectives 51

4.6.2 Expected Process Improvements 52

4.6.3 Financial Interpretation 54

4.7 System Evaluation Results 55

4.7.1 Functional Test Results 55

4.7.2 Evaluation Against Design Targets 56

4.7.3 Limitations of the Findings 57

CHAPTER V CONCLUSION AND RECOMMENDATIONS 58

5.1 Introduction 58

5.2 Summary of the Proposed Study Design and Implementation 58

5.3 Summary of Findings 59

5.4 Conclusion 60

5.5 Recommendation 61

REFERENCES 63

LIST OF TABLES

Table 1. Stakeholder Analysis Matrix 16

Table 2. SIPOC Analysis Framework 17

Table 3. Problems and Root Causes Analysis 19

Table 4. Functional Requirements 21

Table 5. Non-Functional Requirements 23

Table 6. Before-and-After Process Comparison 26

Table 7. Business Logic and Automation Rules Engine 27

Table 8. Sample Sales/Rental Transaction Record 29

Table 9. Project Implementation Phasing 33

Table 10. Risk Assessment and Mitigation Matrix 35

Table 11. Financial Analysis (Costs vs. Benefits) 37

Table 12. Development Tools and Technology Stack 46

Table 13. Hardware Requirements 48

Table 14. Software Requirements 48

Table 15. System Inputs and Outputs 49

Table 16. Study Outputs per Objective 51

Table 17. Current Process Estimates and Design Targets 53

Table 18. Functional Test Cases and Results 55

LIST OF FIGURES

Figure 1. Current State of the Process 18

Figure 2. Fishbone (4P) Root-Cause Diagram 20

Figure 3. Business Analysis Project Framework 24

Figure 4. Proposed Future-State Automated Process 25

Figure 5. ATRIMS-CareMed Three-Tier System Architecture 31

Figure 6. Owner/Admin Operations Dashboard Wireframe 31

Figure 7. Implementation Work Breakdown and Weekly Schedule 34

CHAPTER I

INTRODUCTION

1.1 Background of the Project

Small medical supply enterprises matter a great deal to the Philippine healthcare system. They are the businesses that families turn to when a patient comes home from the hospital and needs a wheelchair, a hospital bed, or an oxygen concentrator to continue recovery at home. CareMed Medical Supply Enterprise is one such business: a small, single-location enterprise that offers durable medical equipment for outright sale and for short-term rental to individual patients, family caregivers, and small clinics in its local service area. Like many micro and small enterprises in the Philippines, CareMed grew by relying on manual, paper-based systems to manage its day-to-day operations.

At present, every sales transaction at CareMed is written by hand into a counter logbook. Rental agreements are filled out on printed forms and their details are later re-entered into an Excel worksheet. Inventory is tracked in that same worksheet, updated only at the end of each day, and often reconstructed from memory, since staff serve customers continuously during operating hours. Customer orders arrive through Facebook Messenger. For refillable oxygen tanks, staff physically count full, rented, and empty cylinders before

every Prime Gas refill run so that consumption can be reconciled against supplier records at the 5th and 20th of each month.

This manual mechanism was tolerable when daily transactions were few. As the enterprise's customer base has grown, driven by increasing home-based care demand, the volume of concurrent sales, rentals, and refills has outgrown what logbooks and end-of-day worksheets can reliably support. Transactions are recorded late or missed entirely when the counter is busy; the worksheet version on the shared computer conflicts with notes kept in Messenger; a rental item that is already committed to one customer is sometimes promised to another because the commitment was never written into the stock record; and monthly reconciliation of sales, rental invoices, and tank refills consumes hours of the owner's time at every cutoff. Periodic physical counts repeatedly reveal differences between recorded and actual stock that cannot be traced to any specific transaction, because no per-transaction audit trail exists.

This project designed a Business Process Automation Plan for an Automated Transaction Recording and Inventory Management System for CareMed (ATRIMS-CareMed). The plan targeted the two operational functions where the enterprise's manual methods fail most severely: the recording of sales and

rental transactions, and the real-time control of inventory. The developed system replaces the logbook-and-end-of-day-worksheet pattern with point-of-transaction digital recording, automatic stock deduction and restoration, rule-based validation, and a complete chronological audit trail, giving the owner, at any moment, an accurate and traceable picture of every item and every transaction.

1.2 Statement of the Problem

1.2.1 General Problem

How can the transaction recording and inventory management processes of CareMed Medical Supply Enterprise be redesigned and automated to eliminate handwritten logbooks and end-of-day worksheet updates, prevent unrecorded transactions and stock discrepancies, and provide real-time, traceable inventory and transaction visibility for the owner and staff?

1.2.2 Specific Problems

1. Delayed and Omitted Transaction Recording: Sales are written into a counter logbook and rentals into separate paper forms, but during peak hours transactions are often recorded late or not at all, forcing staff to reconstruct the day from memory at closing time.
2. No Real-Time Inventory Visibility: Stock levels exist only in an end-of-day Excel worksheet, so staff answering

availability inquiries cannot confirm stock from records and must physically check the storage area; simultaneous transactions can reference the same inventory.

3. Untraceable Stock Discrepancies: Periodic physical counts regularly reveal differences between recorded and actual stock, but with no per-transaction movement log, discrepancies cannot be traced to a specific sale, return, or data-entry error.

4. Double-Commitment of Rental Items: A rental item already committed to an active rental is occasionally promised to a new customer because the commitment was never reflected in the availability record consulted at the counter.

5. Unreliable Refillable Tank Tracking: Full, rented, and empty oxygen tank counts are verified by physical counting before each Prime Gas refill, a process prone to error that directly affects reconciliation against supplier records at each cutoff.

6. Labor-Intensive Reporting: Compiling monthly sales, active rental, and inventory summaries requires manually consolidating logbooks, worksheets, and chat histories, a process that takes hours and reflects only the state of the last manual update.

1.3 Objectives of the Project

1.3.1 General Objective

To design an automated Business Process Automation (BPA) Plan and functional system architecture for ATRIMS-CareMed that replaces handwritten logbooks and end-of-day worksheet updates with point-of-transaction digital recording, enforces real-time inventory control with a complete audit trail, and automates validation and reporting for CareMed Medical Supply Enterprise.

1.3.2Specific Objectives

1. Analyze and document the current manual transaction recording and inventory process using business analysis methodologies (SIPOC, As-Is Process Mapping, and 4P Root-Cause Analysis).
2. Identify specific process gaps, bottlenecks, and data risks in the current logbook-and-worksheet mechanism.
3. Perform a full Stakeholder Analysis to capture operational requirements across the owner, counter staff, inventory custodian, delivery personnel, and customers.
4. Design an improved Future-State (To-Be) business workflow incorporating point-of-transaction recording, real-time stock deduction and restoration, rule-based validations, and automated alerts.

5. Specify the functional and non-functional system requirements for the proposed web-based ATRIMS-CareMed platform.
6. Formulate automated business rules for inventory control, including low-stock thresholds, rental item locking, and oxygen tank refill statuses.
7. Construct a functional prototype and interactive dashboard wireframe to validate operational usability for the admin side of the business.
8. Develop a test plan containing functional validation test cases for core transaction and inventory processing scenarios.
9. Formulate a risk management strategy, implementation roadmap, cost-benefit justification, and change management plan.

1.4 Significance of the Project

For the Owner/Admin: Provides a real-time dashboard of stock on hand, items rented out, and daily sales, replacing hours of end-of-day reconciliation with instant, exportable reports, and establishing a complete audit trail for every transaction.

For Counter and Sales Staff: Eliminates double entry between logbook and worksheet; a sale recorded once automatically updates stock, computes totals (including rental

deposits based on delivery distance), and produces a transaction reference number.

For the Inventory Custodian: Every stock movement (sale, rental checkout, rental return, refill, or adjustment) is logged with a timestamp and user identifier, making periodic counts reconcilable and discrepancies traceable to a specific transaction.

For Delivery Personnel: Rental commitments are visible before dispatch, preventing trips for items that are no longer available, and oxygen tank refill quantities are computed from system records rather than physical re-counts.

For Customers: Accurate availability answers at the counter and documented transaction references instead of informal chat-based records.

For Future Business Analysis Researchers: Demonstrates a repeatable framework for digitizing transaction recording and inventory control in micro and small medical supply enterprises with limited capital and no dedicated IT staff.

1.5 Purpose and Description

The primary deliverable of this project is a Business Process Automation Plan that establishes the functional specifications, architectural framework, and implementation strategy for ATRIMS-CareMed, targeted at the admin-side

operations of CareMed: recording transactions and controlling inventory.

The platform's core functional capabilities are organized into five pillars: (1) Point-of-Transaction Recording: one-screen capture of sales and rentals with automatic totals and reference numbers; (2) Real-Time Inventory Ledger: immediate stock deduction, restoration, and status tracking for every item including refillable tanks; (3) Rules Engine: balance and stock validations, low-stock thresholds, and rental locking; (4) Notifications and Alerts: low-stock and overdue-rental alerts on the admin dashboard; and (5) Audit and Analytics: chronological inventory and transaction logs, searchable history, and exportable CSV reports for every cutoff period.

Through this platform, the admin records a sale or rental in a single screen, and the system does the rest: it validates stock, deducts or locks inventory, generates a transaction reference number, updates all dashboards instantly, and writes an immutable audit entry. Every end-of-day reconciliation task the owner previously performed by hand is now replaced by data the system has already captured at the moment each transaction occurred.

1.6 Scope and Limitations

Scope:

1. Admin-side transaction recording for sales and rentals, including deposit calculation based on delivery distance.
2. Real-time inventory ledger for all products, with statuses for refillable items (Full / Rented / Empty-For-Refill) supporting the oxygen tank workflow.
3. Automated stock deduction on confirmed transactions and automatic restoration on rental returns and refills.
4. Low-stock threshold alerts and reorder-level reporting per product.
5. Chronological inventory movement log and transaction audit trail with timestamps and user identifiers, searchable and exportable (CSV/Excel).
6. Owner/admin dashboard showing daily sales, active rentals, low-stock items, and tank refill counts.
7. Limitations:
8. Customer Self-Service Portal: Out of scope for this plan; customers interact with staff at the counter or via existing Messenger channels, and the system generates references staff relay to them.
9. Native Mobile Application: The platform is web-based and mobile-responsive but does not include a native iOS/Android app or push notifications.

10. Barcode/QR Scanning: Item identification is manual or by search; hardware scanning is identified as a future enhancement.
11. Accounting and Payroll: The system produces transaction and inventory reports but does not perform full bookkeeping, tax filing, or payroll processing.

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

2.1 Related Literature

2.1.1 The automation of business processes in small and medium-sized enterprises

Business Process Automation (BPA) uses digital tools to improve repeated business activities and make work more consistent. García-Vidal et al. (2025) observed that data-driven process management can help MSMEs improve operations and decision-making, although the availability of resources and technical capability can affect implementation [1]. Menezes (2023) also emphasized that automation should begin with a clear understanding of the existing process and its requirements [2]. These ideas apply to CareMed because its sales, rental, and inventory-handling activities had been carried out through separate manual records.

2.1.2 Digital records and the quality of information

Digital transaction recording can improve the timeliness and consistency of business information. Ghalavand et al. (2024) identified accuracy, completeness, consistency, timeliness, and accessibility as important dimensions of data quality in health information systems [3]. Niyomugaba et al. (2026)

likewise stated that electronic logistics systems can improve stock monitoring and data visibility, although inaccurate data and missing alerts can still reduce their benefits [4]. For CareMed, these findings indicate that sales and rentals should be recorded when they take place, that stock should be updated right away, and that all information should be kept in a single system that is easy to access.

2.1.3 The management of medical inventory and rentals

Medical inventory requires reliable information because unavailable or incorrectly recorded items can affect service delivery. Xu, Wermus, and Bauman (2011) demonstrated the value of incorporating medical supply information and inventory data into a single information system [5]. Johnson and Ebakivie (2026) likewise found that technology can support healthcare inventory tracking and stock management [6]. Tracking is also required for items on rent, including their status and movement. Since an item is unavailable while it is being rented, Odoo, for example, connects rental orders with product pickup, return, scheduling, and inventory movement [7]. These practices support the inclusion of real-time stock updates, rental locking, and oxygen-tank status tracking in ATRIMS-CareMed.

2.1.4 Audit Trails and Requirements Analysis

Audit trails keep a record of every action taken in a system. They help track what happened, when it happened, and who did it. This makes it easy to spot mistakes or find out if someone tried to do something wrong. Teams can use audit trails to look at past events, fix problems, and make better choices in the future.

Requirements analysis is the process of figuring out what a project needs. It involves talking to users, gathering information, and writing down clear goals. Good requirements analysis helps make sure the final product meets the needs of everyone involved. It also helps avoid confusion and reduces the chance of mistakes during the project.

A reliable transaction system should likewise maintain a clear record of major changes. The third revision of NIST SP 800-171 contains audit and accountability controls that stress recording the details relevant to traceability [8]; this supports the use of timestamps, user identifiers, and transaction references in ATRIMS-CareMed. Furthermore, the IIBA Business Analysis Standard recognizes requirements and traceability as important in connecting stakeholder needs to solution requirements [9]. This is consistent with the

methodology of this study, in which stakeholders were first identified and their needs were then converted into system requirements.

2.2 Related Systems

Existing systems show that many of the functions required by CareMed are already available in commercial software, although each program is built around different business requirements. Square connects sales with inventory updates [10], while Zoho Inventory provides transaction history and inventory adjustments [11]. Odoo integrates its inventory and rental functions [7]. These systems provide useful references, but CareMed requires a specific combination of sales, medical-equipment rentals, oxygen-tank tracking, rental availability, and reporting.

2.3 Synthesis

The reviewed literature shows that automation is most useful when it is based on a clear understanding of the existing process. Digital systems can improve the visibility of business processes and information, reduce delays, and support better tracking, while rental systems show the need to manage item status from checkout to return. Although some of these functions are offered by existing platforms, CareMed needs them combined with its own rules for medical-equipment rentals, oxygen-tank status, stock alerts, audit history, and

reporting. ATRIMS-CareMed was therefore developed as a focused system that addresses the specific process gaps identified in Chapter I and provides a more consistent method of recording transactions and managing inventory.

CHAPTER III

METHODOLOGY, RESULTS, AND DISCUSSION

3.1 Requirements Analysis

3.1.1 Stakeholder Analysis

Stakeholders are the people affected by the system. Table 1 lists who they are and what they need.

Table 1. Stakeholder Analysis Matrix

Stakeholder	Organizational Role	Core Responsibilities	Primary Pain Points / Needs	System Impact / Expectations
Owner / Administrator	Business owner	Approves transactions and checks reports	Hours spent reconciling records	Real-time dashboard and reports
Counter / Sales Staff	Front line	Records sales and rentals	Double entry and forgotten records	Simple one-screen recording
Inventory Custodian	Stock controller	Manages storage and counts	Counts do not match records	Stock movement history
Delivery Personnel	Fulfilment support	Delivers and picks up items	Trips for unavailable items	See availability before delivery
Customer	End recipient	Buys or rents equipment	Item said to be available is not	Correct availability and a reference number
Supplier (Prime Gas)	External supplier	Refills oxygen tanks	Disputed refill counts	Clear tank counts per cutoff

3.1.2 SIPOC Analysis

SIPOC gives a simple big-picture view of the process.

Table 2. SIPOC Analysis Framework

Supplier	Input	Process	Output	Customer
Customer / Counter Staff	Order details	Record Transaction	Recorded transaction	Owner / Admin
ATRIMS Rules Engine	Product and stock list	Validate and Post	Updated stock	Inventory Ledger
Inventory Custodian	Count results and returns	Update Stock	Adjusted stock	Counter Staff / Owner
ATRIMS Reporting Module	Transaction and stock data	Generate Reports	Reports and dashboard	Owner / Supplier Reconciliation

3.1.3 Current-State Process

Today, sales are written in a logbook and stock is updated in Excel at the end of the day. Rentals are noted on paper forms and in Messenger.

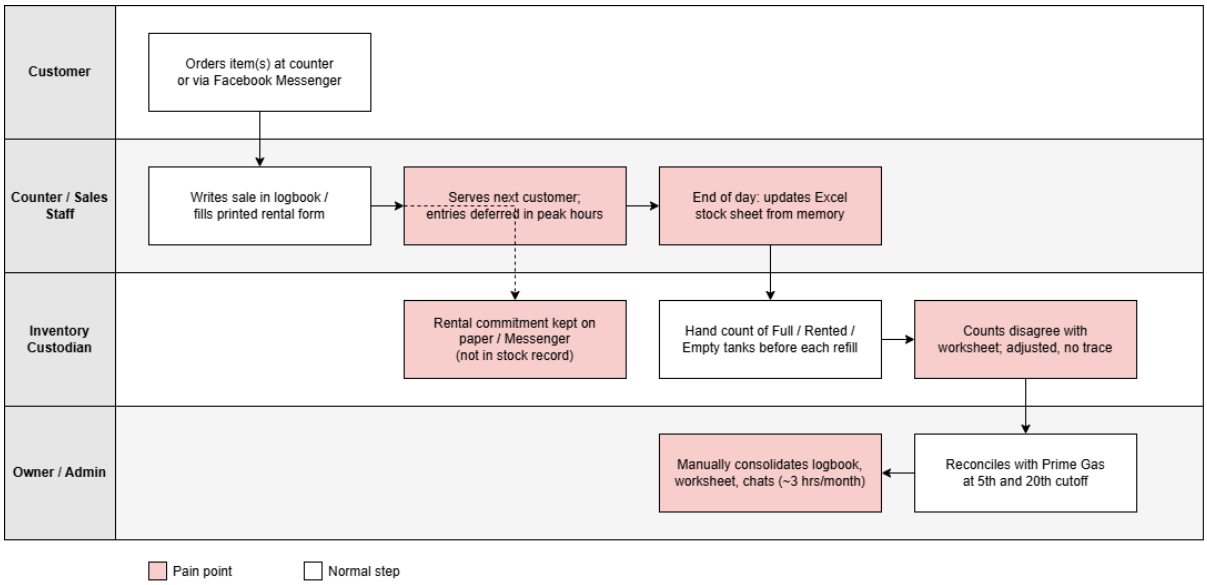


Figure 1. Current State of the Process

3.1.4 Problems Identified

Through field interviews and root-cause analysis, key operational bottlenecks were identified:

Table 3. Problems and Root Causes Analysis

Problem Identified	Root Cause	Operational Impact
1. Late or Omitted Transaction Recording	Staff record only when free	Some transactions are late or missed
2. Persistent Stock Discrepancies	Stock updated only at end of day	Chances that counted items do not match records
3. Double-Committed Rental Items	Rentals are not in the stock record	An item may be promised twice
4. Unreliable Tank Refill Counts	Tanks are counted by hand	Wrong refill quantities
5. Multi-Hour Report Compilation	Records are in many places	Reports take about 3 hours

3.1.5 Root-Cause Analysis (Fishbone 4P Diagnostic)

The 4P Fishbone examined the causes of the problems under People, Process, Platform, and Paperwork.

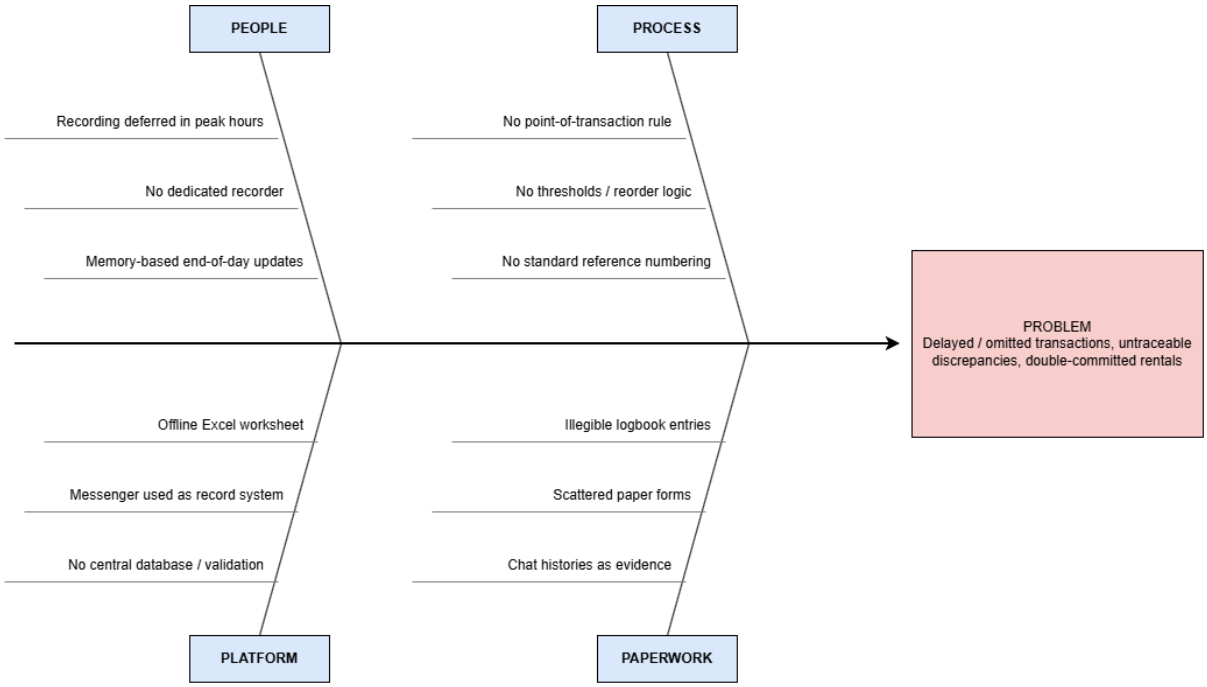


Figure 2. Fishbone (4P) Root-Cause Diagram

People: Counter staff serve customers continuously and record transactions only when time permits; at closing, the day's entries are reconstructed from memory. No role is accountable for the timeliness or completeness of the record.

Process: There is no rule that a transaction must be recorded at the moment it occurs; no automatic stock consequence when one is recorded; no reference numbering standard; and no reconciliation step between physical counts and recorded balances.

Platform (Technology): The worksheet is offline, shared, and editable by anyone; Messenger is used as a transactional record store without any structure; there is no central database, no validation, and no audit log.

Paperwork (Information): Logbook entries are handwritten and occasionally illegible; rental terms on paper forms are not linked to stock records.

Core Diagnostic Finding: The primary problem is not staff carelessness but the reliance on a deferred, memory-based recording mechanism with no real-time validation, no enforced point-of-capture rule, and no traceability: a structural defect that worsens as transaction volume rises.

3.1.6 Requirements Engineering

Functional requirements define the core actions and system capabilities that ATRIMS-CareMed must perform.

Table 4. Functional Requirements

ID	Requirement	Stakeholder	Priority
FR01	Keep a real-time inventory record per product (quantity, rental/refill status).	Admin / Custodian	High
FR02	Record a sale on one screen: auto-compute totals, distance-based rental deposit,	Counter Staff	High

ID	Requirement	Stakeholder	Priority
	and a unique reference number.		
FR03	Deduct stock on confirmed sale; mark stock as Rented on rental checkout.	System	High
FR04	Block any transaction where the quantity requested exceeds available stock.	System	High
FR05	Restore stock on rental return; send refillable items to Empty-For-Refill.	System	High
FR06	Track tank status (Full / Rented / Empty-For-Refill) and compute refill quantities for supplier reconciliation.	Admin / Supplier	High
FR07	Alert when stock reaches or falls below a per-product reorder level.	Admin	Medium
FR08	Log every stock change with timestamp, user, transaction reference, and reason.	Admin / Custodian	High
FR09	Search transaction history by reference number, customer, date, or product.	Admin / Staff	High
FR10	Export CSV/Excel reports per cutoff period: sales, active rentals, low stock, tank status.	Admin	High

Table 5. Non-Functional Requirements

Category	Target	Why It Matters
Usability	Mobile-friendly screens; first-time user can record a sale in under 2 minutes.	Staff mainly use smartphones and one shared computer.
Performance	Transactions and stock updates post in under 2 seconds.	Keeps counter service fast at peak hours.
Availability	99.5% uptime in operating hours; always-online operation with a prepaid 5G pocket Wi-Fi failover at the counter.	A backup connection keeps every stock check live, so rentals are never confirmed without real-time validation.
Reliability & Audit	Daily automatic backups; no data loss; permanent audit log of transactions and stock movements.	Supports cutoff reconciliation and dispute resolution.
Maintainability	Admin can edit thresholds, deposit rules, and product data without code changes.	Owner can adjust as suppliers and delivery costs change.

3.2 Framework of the Project

The overall project methodology follows an 11-stage Business Analysis Project Framework, moving from organizational problem definition to continuous system improvement.

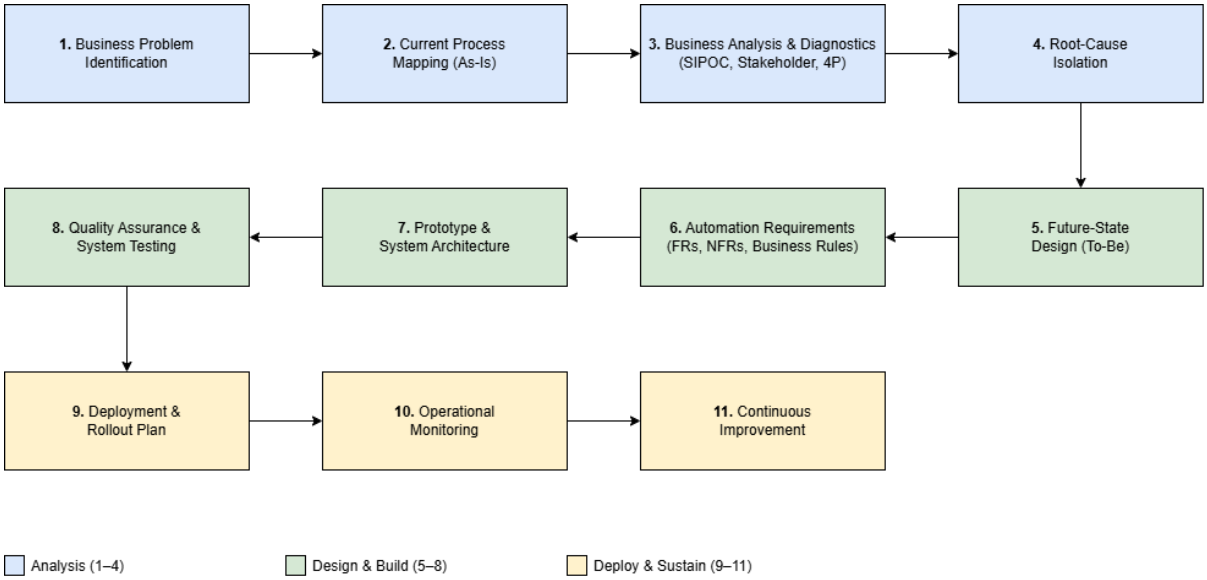


Figure 3. Business Analysis Project Framework

3.2.1 Proposed Future-State Process

The proposed Future-State process requires recording at the moment of the transaction, applies checks automatically, and keeps the inventory ledger accurate and traceable at all times.

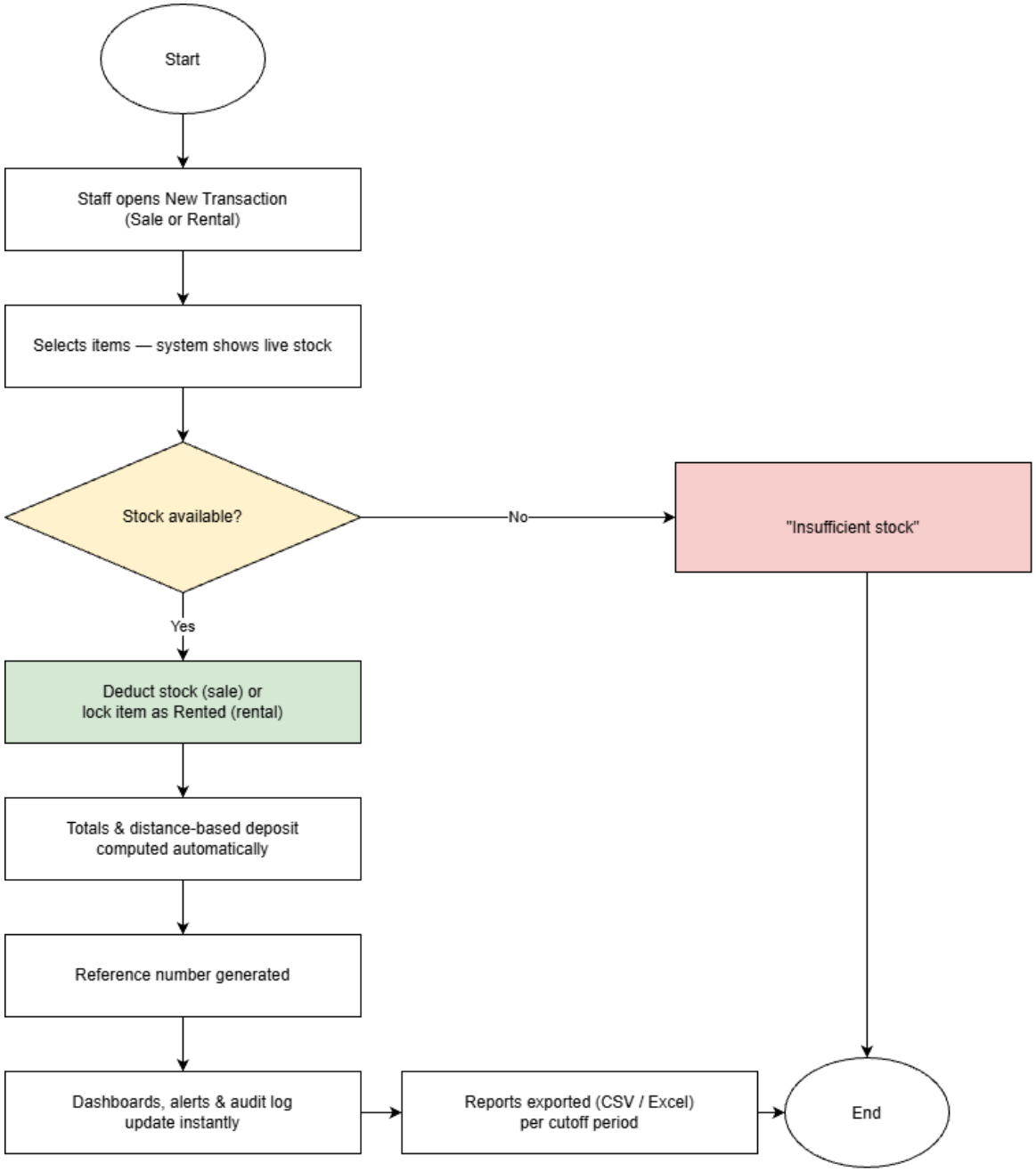


Figure 4. Proposed Future-State Automated Process

3.2.2 Before and After Comparison

Table 6. Before-and-After Process Comparison

Operational Dimension	Current Manual Process (As-Is)	Proposed Automated Process (ATRIMS-CareMed)
Recording Moment	Handwritten logbook entry, often put off until the end of the day and written from memory.	Digital capture at the time of the transaction; a sale cannot proceed without a record.
Stock Updates	End-of-day worksheet update; discrepancies are found only during physical counts.	Instant deduction on sale, locking on rental, and restoration on return; always up to date.
Rental Commitments	Paper forms and Messenger notes; not shown in the availability record.	Rented items are locked in the ledger, so availability answers can be trusted.
Tank Refill Counts	Hand count of full/rented/empty tanks before each refill.	System-kept statuses; refill quantity computed per cutoff.
Discrepancy Tracing	Not possible; there is no per-transaction trail.	Every movement is traced to a user, timestamp, and transaction reference.
Reporting	About 3 hours of manual compiling per month.	One-click exportable reports from live data in under 5 minutes.

3.2.3 Proposed System Modules

ATRIMS-CareMed is made up of four modules:

1. Module 1 – Real-Time Inventory Ledger: Keeps the current stock per product, rental locks, refillable tank statuses (Full / Rented / Empty-For-Refill), reorder thresholds, and the dated movement log.
2. Module 2 – Transaction Recording Engine: The admin-side core. It records sales and rentals on a single screen with automatic totals, distance-based deposits, and reference numbers.
3. Module 3 – Alerts & Notifications Engine: Raises low-stock alerts, flags overdue rentals, and computes tank refill quantities for each supplier cutoff.
4. Module 4 – Reporting & Audit: Owner dashboards, searchable transaction history, and exportable CSV/Excel reports for sales, rentals, inventory, and tank status.

3.2.4 Automation Rules Engine

To apply the rules the same way every time and remove guesswork, ATRIMS-CareMed runs on a business rules engine:

Table 7. Business Logic and Automation Rules Engine

Rule ID	Operational Trigger / Condition	Automated Action Executed by System
RULE-01	Requested quantity is more than the available stock for any line item.	Block the transaction and show: "Insufficient stock for [item]."
RULE-02	Sale transaction confirmed.	Deduct line quantities from the inventory ledger and write a movement-log entry.

Rule ID	Operational Trigger / Condition	Automated Action Executed by System
RULE-03	Rental transaction confirmed.	Lock the item as Rented (unavailable for sale or other rentals) and record the deposit based on delivery distance.
RULE-04	Rental return recorded by custodian.	Restore the item to available stock; for refillable tanks, set status to Empty-For-Refill.
RULE-05	Available quantity is at or below the reorder threshold of a product.	Raise a low-stock alert on the admin dashboard and add the item to the reorder report.
RULE-06	Physical count differs from recorded stock.	Require an adjustment with a reason code and admin approval; log it as an adjustment entry.
RULE-07	Void/cancel of a recorded transaction.	Reverse the stock effects, require a reason, and record the void in the audit log.
RULE-08	Rental return recorded with a returned quantity lower than the quantity rented (partial return).	Restore only the returned quantity to available stock; keep the unreturned balance locked as Rented; keep the rental record open and flag the outstanding quantity on the dashboard.
RULE-09	Rental return recorded after the agreed due date (overdue).	Compute the late fee as days overdue multiplied by the daily late rate set by the Owner/Admin; add it to the balance due and write it to the audit log.
RULE-10	Item returned damaged, as marked by the custodian during return.	Require a damage reason code; set the item to Damaged (unavailable for sale or rental); assess the damage fee against the deposit under the Owner/Admin schedule; log it as an adjustment entry with admin approval.

3.2.5 Sample Sales Transaction Record

The operational lifecycle of a recorded transaction within ATRIMS-CareMed is illustrated by the following record:

Table 8. Sample Sales/Rental Transaction Record

Data Attribute Field	Production System Sample Value	Attribute Operational Description
Transaction ID	TRX-2026-00341	Unique system-generated tracking key.
Transaction Type	Rental (with delivery)	Sale or rental classification with fulfilment mode.
Staff Recorder	Staff account: R. Villanueva	User who recorded the transaction (audit attribution).
Customer	Teresita L. Ocampo (walk-in)	Customer served at the counter.
Item / Product	Oxygen Concentrator (OC-5L)	Rented equipment referenced to the inventory ledger.
Rental Period	Oct 06 – Nov 05, 2026	One-month rental billing period.
Monthly Rate	₱3,500.00	Configured rental price for the item.
Distance-Based Deposit	₱1,500.00	Deposit computed from delivery distance rule.
Stock Effect	1 unit locked → Rented	Item removed from availability upon confirmation (RULE-03).
Recorded At	Oct 06, 2026 | 10:42:18 AM	Point-of-transaction timestamp.
Audit Log Ref	MOV-2026-01877	Movement-log entry linking stock change to this transaction.

3.2.6 Proposed Dashboard and Visual Layout

The platform follows a three-tier architecture: a presentation layer of mobile-responsive admin screens, an application layer hosting the rules engine and workflow logic,

and a data layer holding the central ledger, movement logs, and audit trail, with automated daily backups.

The system is built as a custom web application using Python 3 with the Flask framework for the application layer and rules engine, Bootstrap 5 for the mobile-responsive presentation layer, and MySQL accessed through the SQLAlchemy ORM for the data layer. It is packaged with Docker, served through Gunicorn, and planned to be hosted on a DigitalOcean Droplet at USD 6 per month. MySQL (InnoDB) transactions with row-level locking let the system lock a rental item at the moment of confirmation, so two staff members cannot check out the same item at once. The application runs online only, with a pocket Wi-Fi failover at the counter, which keeps every stock check against the central ledger and keeps the build within the ₱100,000 development budget.

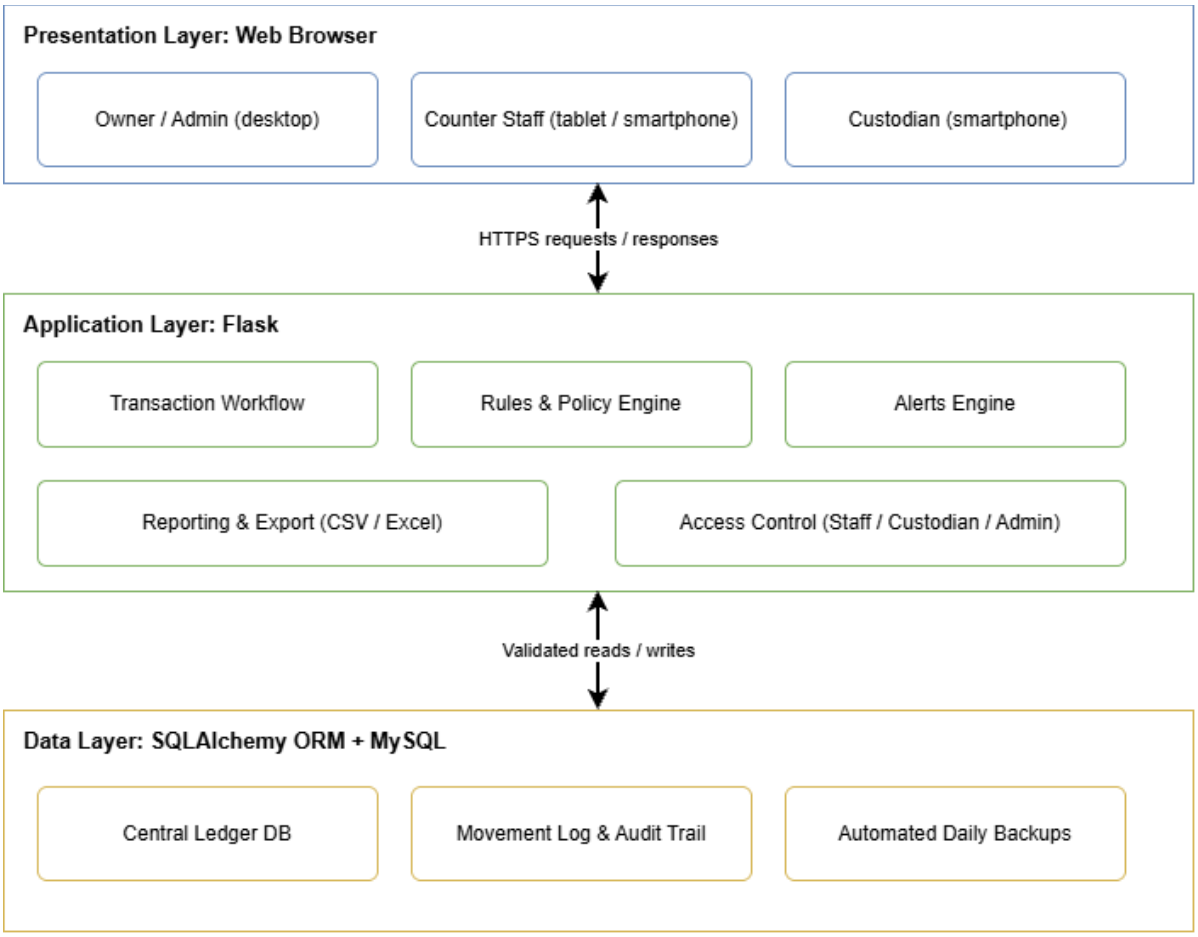


Figure 5. ATRIMS-CareMed Three-Tier System Architecture

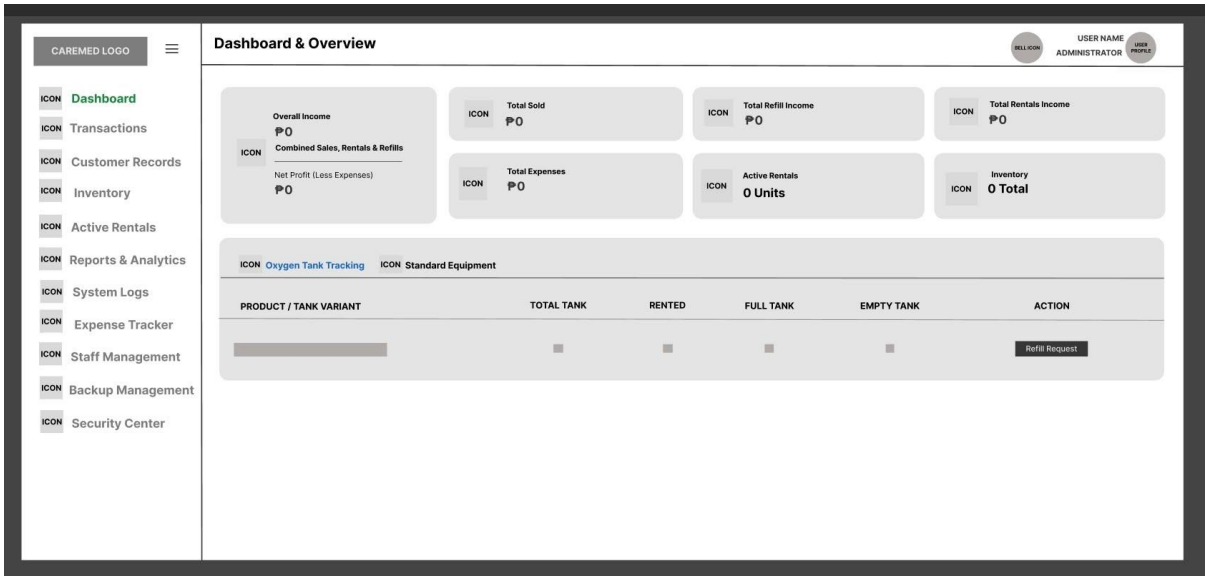


Figure 6. Owner/Admin Operations Dashboard Wireframe

3.3 Development and Testing

3.3.1 Prototype Layouts

The functional prototype includes three responsive admin-side views:

Screen 1 – Operations Dashboard: The owner's overview — today's sales, transactions recorded, low-stock items, and tank status summary.

Screen 2 – New Transaction Screen: The single-screen recorder — item search with live stock display, sale/rental toggle, quantity, delivery distance for deposit computation, and instant validation messages.

Screen 3 – Inventory & Reconciliation Workspace: Stock ledger with movement history per item, physical-count adjustment entry with reason codes, tank status board, and the export panel for cutoff reports.

3.4 Implementation Plan

3.4.1 Project Phases

Table 9. Project Implementation Phasing

Phase Sequence	Major Phase Milestone	Key Operational Activities	Deliverables Produced
Phase 1	Project Initiation & Requirements	Stakeholder sign-offs, policy reviews, requirements baselining with the owner.	Approved Requirements Document & Project Charter.
Phase 2	Architecture & UI Design	Database design, wireframe prototyping, validation rules and threshold configuration.	System Architecture Blueprint & Interactive Wireframes.
Phase 3	Core System Development	Ledger, transaction engine, rules engine, alerts, and report exports.	Fully Functional ATRIMS-CareMed Release Candidate.
Phase 4	Verification & User Acceptance	Execution of test suites; parallel-run pilot alongside logbook for two weeks.	UAT Sign-off Certificate & System Defect Log.
Phase 5	Deployment & Training	Production setup, staff training, opening balance migration, go-live.	Production Launch at CareMed premises.
Phase 6	Operational Governance	Ongoing KPI tracking, backup verification, quarterly rule reviews.	Post-Implementation Review & SLA Metrics Report.

3.4.2 Implementation Schedule (Gantt Representation)

Implementation Activity & Description	W1	W2	W3	W4	W5	W6	W7	W8
1. Requirements Finalization & Policy Mapping								
2. Database Schema & Architecture Design								
3. UI Prototype Development & Owner Review								
4. Core Ledger & Transaction Engine Development								
5. Rules Engine, Alerts & Report Exports								
6. UAT & Two-Week Parallel Pilot Run								
7. Staff Training & Opening Balance Migration								
8. Production Go-Live (Full Cutover)								
9. Post-Launch Review & KPI Audit								

Figure 7. Implementation Work Breakdown and Weekly Schedule

3.4.3 Risk Analysis and Mitigation Matrix

Table 10. Risk Assessment and Mitigation Matrix

Identified Risk Description	Risk Severity	Likelihood	Impact	Practical Mitigation Strategy
1. Staff Resistance to Digital Recording	High	Medium	High	Keep the recorder to one screen; run a two-week parallel pilot; involve staff in screen design.
2. Unstable Internet at the Premises	Medium	High	Medium	Prepaid 5G pocket Wi-Fi at the counter as automatic failover; system stays online-only so stock is always validated in real time; no rental is confirmed while disconnected; daily backup.
3. Wrong Opening Balances at Migration	High	Medium	High	Physical count witnessed by owner and custodian; two-stage verification before cutover.
4. Data Loss	High	Low	High	Daily automated backups.
5. Device Unavailability (single shared computer)	Medium	Medium	Medium	Any-device browser access, including staff smartphones; no client software required.

3.4.4 Cost-Benefit Analysis and ROI Calculation

The cost-benefit analysis evaluates the estimated expenses required to implement ATRIMS-CareMed and the potential financial benefits of automating CareMed Medical Supply Enterprise's transaction recording and inventory management processes. The system includes sales and rental recording, real-time inventory updates, rental availability controls, oxygen tank tracking, audit trails, automated reports, and administrative dashboards. Since ATRIMS-CareMed is a custom web-based application, its estimated implementation cost includes software development, staff training, documentation, and cloud hosting with backups. The following estimates are intended for project planning and should be validated through actual supplier quotations or developer estimates before implementation.

Table 11. Financial Analysis (Costs vs. Benefits)

Cost Classification	**Estimated Budget (PHP)**	**Financial and Operational Details**
Custom Software Development	₱100,000	Development of the backend, user interface, database, transaction recording, inventory management, automation rules, and reporting modules
Staff Training and Documentation	₱8,000	Staff orientation, hands-on training, user guides, and quick-reference materials
Cloud Hosting and Backups (Year 1)	₱5,889	Estimated cloud hosting expenses for the first year, including an allowance for data backups
Subtotal	**₱113,889**	**Estimated implementation and first-year hosting expenses**
Contingency Reserve (10%)	₱11,389	Allowance for unexpected expenses and minor project adjustments
TOTAL ESTIMATED FIRST-YEAR COST	**₱125,278**	**Approximately ₱125,300**

The cost estimates cover the development of ATRIMS-CareMed, staff training and documentation, and first-year cloud hosting and backups. The subtotal is ₱113,889, with

an additional contingency reserve of 10%, equivalent to ₱11,389. This results in a total estimated first-year cost of ₱125,278. The contingency reserve provides financial flexibility for unforeseen expenses and minor project adjustments during implementation.

Estimated Annual Operational Benefits

The expected benefits of ATRIMS-CareMed include reduced administrative workload, fewer inventory discrepancies, more efficient rental monitoring, and faster report preparation. These improvements may reduce operating expenses and allow personnel to devote more time to customer service and inventory control.

For preliminary financial evaluation, the analysis assumes approximately 100 transactions per week, equivalent to 5,200 transactions annually. It also assumes that manual transaction processing requires approximately 10 minutes per transaction, while the automated process reduces the required effort to approximately 2 minutes per transaction. These assumptions must be validated through actual observations or time measurements at CareMed Medical Supply Enterprise.

Using the assumed administrative labor rate of ₱80 per hour, the estimated annual processing costs are calculated as follows:

Annual Manual Processing Hours = 5,200 transactions × 10 minutes ÷ 60 = 866.67 hours

Annual Automated Processing Hours = 5,200 transactions × 2 minutes ÷ 60 = 173.33 hours

Annual Manual Processing Cost = 866.67 hours × ₱80 = ₱69,333

Annual Automated Processing Cost = 173.33 hours × ₱80 = ₱13,867

Estimated Annual Direct Labor Savings = ₱69,333 – ₱13,867 = ₱55,466

The estimated direct labor savings represent the potential value of the time saved through automated transaction processing. These savings do not necessarily represent an actual reduction in payroll expenses, as personnel may continue receiving the same salaries while using the time saved for other responsibilities. Additional potential benefits may arise from fewer inventory discrepancies, reduced rental conflicts, less time spent preparing reports, and fewer unnecessary supplier reconciliation trips. However, these benefits should be supported by business records or clearly

identified as preliminary estimates rather than guaranteed savings.

For the initial ROI calculation, the projected annual financial savings of ₱133,968 are retained as a preliminary assumption. This amount represents the combined estimated direct labor savings and other potential operational savings. The additional savings beyond direct labor must be independently validated before the total can be treated as a reliable financial projection.

Return on Investment (ROI) Calculation

The return on investment measures the potential financial benefit of ATRIMS-CareMed relative to its estimated first-year cost. Using the revised estimated first-year cost of ₱125,278 and the preliminary projected annual financial savings of ₱133,968, the calculations are as follows:

Net Financial Benefit (Year 1) = Annual Financial Savings – Total First-Year Cost

Net Financial Benefit (Year 1) = ₱133,968 – ₱125,278

Net Financial Benefit (Year 1) = ₱8,690

The return on investment is calculated using the following formula:

ROI = (Net Financial Benefit ÷ Total First-Year Cost) × 100%

ROI = (₱8,690 ÷ ₱125,278) × 100%

ROI ≈ 6.94%

The estimated payback period indicates how long it may take for the projected financial savings to recover the initial investment.

Payback Period = (Total First-Year Cost ÷ Annual Financial Savings) × 12 months

Payback Period = (₱125,278 ÷ ₱133,968) × 12 months

Estimated Payback Period ≈ 11.2 months

Based on these preliminary assumptions, ATRIMS-CareMed may recover its estimated first-year cost within approximately 11.2 months and generate a net financial benefit of ₱8,690 during the first year. The projected first-year ROI is approximately 6.94%, provided that the assumed annual savings are achieved and the estimated costs remain accurate.

Conclusion

The cost-benefit analysis indicates that ATRIMS-CareMed has an estimated first-year cost of ₱125,278, including software development, staff training and documentation, cloud hosting and backups, and a 10% contingency reserve. Compared with the preliminary projected annual financial

savings of ₱133,968, the system may generate a positive net financial benefit of ₱8,690 during the first year. However, these results depend on the accuracy of the estimated development cost, transaction volume, processing time, labor rate, and additional operational savings. The projected ROI and payback period should therefore be interpreted as preliminary estimates rather than guaranteed financial outcomes. Actual implementation expenses and operational improvements should be measured to determine the system's financial viability.

In addition to potential financial savings, ATRIMS-CareMed may provide operational benefits through improved inventory accuracy, more reliable transaction records, better rental availability monitoring, and faster report generation. These benefits can contribute to more organized business operations and support better decision-making at CareMed Medical Supply Enterprise.

3.4.5 Change Management and Training Framework

To support organizational adoption, a change management program based on the ADKAR Model (Awareness, Desire, Knowledge, Ability, Reinforcement) will be implemented. Awareness: a short briefing with the owner and staff on why logbooks and end-of-day worksheets are

being replaced. Desire: highlight the direct benefits to each role — no more end-of-day re-keying for staff, instant stock visibility for the owner. Knowledge: role-specific hands-on training on the three admin screens. Ability: a two-week parallel pilot in which the system will run alongside the logbook, building confidence before cutover. Reinforcement: the logbook is retired on a fixed date, and the first monthly report is generated from the system in front of the staff.

Tailored Training Curricula: Counter/Sales Staff (60 minutes) — recording sales and rentals, deposits, and handling stock-block messages. Inventory Custodian (45 minutes) — returns, refills, tank statuses, and physical-count adjustments. Owner/Admin (half-day workshop) — thresholds and rules configuration, report exports, audit-log review, and backup verification.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the outputs of the study: the structure of ATRIMS-CareMed, its tools and requirements, how the study objectives were addressed, the improvements the design is expected to bring, and the results of the functional tests. All 13 functional test cases passed, but live use at CareMed has not been measured, so the improvements and financial figures in this chapter remain design targets and preliminary estimates, not measured results.

4.1 Implementation

ATRIMS-CareMed is a custom web-based application that replaces the paper logbook and end-of-day Excel worksheet with point-of-transaction recording and a real-time inventory ledger. It follows the three-tier architecture described in Chapter III.

4.1.1 Presentation Layer

The presentation layer is a mobile-responsive web interface built with HTML5, CSS3, and Bootstrap 5. It runs in a standard web browser, so no software has to be installed on the counter computer or on staff smartphones. It has three admin-side screens:

1. Operations Dashboard: Shows daily sales, transactions recorded, low-stock items, and a summary of oxygen tank status for the owner.

2. New Transaction Screen: A single screen for recording sales and rentals, with item search, live stock display, a sale/rental toggle, delivery distance for the deposit computation, and instant validation messages.

3. Inventory & Reconciliation Workspace: A stock ledger with movement history per item, physical-count adjustments with reason codes, a tank status board, and report exports.

4.1.2 Application Layer

The application layer is written in Python 3 using the Flask framework. It contains the business logic and four modules: the Real-Time Inventory Ledger, the Transaction Recording Engine, the Alerts & Notifications Engine, and Reporting & Audit. The Automation Rules Engine (RULE-01 to RULE-10, listed in Table 7 of Chapter III) applies the stock, rental, return, adjustment, and fee rules in the same way for every transaction.

4.1.3 Data Layer

The data layer uses a MySQL database accessed through the SQLAlchemy ORM. Database transactions with row-level locking prevent two staff members from checking

out the same rental item at the same time. Every stock change is written to a movement log, and the database is backed up automatically every day.

4.1.4 Deployment Setup

The application is packaged with Docker, served through Gunicorn, and planned to be hosted on a DigitalOcean Droplet at USD 6 per month. It runs online only. A prepaid 5G pocket Wi-Fi unit at the counter serves as a backup connection, so every stock check is made against the central ledger.

4.2 Development Tools

Table 12. Development Tools and Technology Stack

Tool / Technology	Category	Purpose
Python 3	Programming language	Backend logic and rules enforcement.
Flask	Web framework	Routing, request handling, and page templates.
Bootstrap 5	Front-end framework	Mobile-responsive screens.
SQLAlchemy	ORM	Database access and schema management.
MySQL (InnoDB)	Database	Transaction storage with row-level locking.

Tool / Technology	**Category**	**Purpose**
Docker	Containerization	Same runtime across development and production.
Gunicorn	WSGI server	Serves the application in production.
DigitalOcean	Cloud hosting	Virtual private server for the application and database.
draw.io	Diagramming	Process flows, architecture diagram, and Gantt chart.
Visual Studio Code	Code editor	Writing and debugging the application.
Git and GitHub	Version control	Tracking code changes.

4.3 Hardware Requirements

The system needs little hardware beyond what CareMed already has.

Table 13. Hardware Requirements

Component	Requirement
Cloud server	1 vCPU, 1 GB RAM, and 25 GB SSD storage (entry-level virtual private server).
Counter terminals	A shared PC or smartphone with a modern web browser.
Network	A stable internet connection, with a prepaid 5G pocket Wi-Fi unit as backup.

4.4 Software Requirements

Table 14. Software Requirements

Environment	Software	Purpose
Server	Linux (Ubuntu LTS) with Docker	Runs the application and database containers.
Server	Python 3, Flask, Gunicorn, MySQL	Application logic, web serving, and data storage.
Counter terminals	Modern web browser (Chrome, Edge, or Safari)	Opens the web screens; no installation needed.
Development	Visual Studio Code, Git, draw.io	Coding, version tracking, and diagrams.

4.5 Inputs and Outputs of the Study

Table 15 shows the main inputs the system takes for each administrative process and the outputs it produces.

Table 15. System Inputs and Outputs

Process	Inputs	System Processing	Outputs
Sale or rental recording	Product, sale or rental, quantity, customer name, delivery distance.	Checks stock (RULE-01), computes totals and deposit, deducts or locks stock (RULE-02, RULE-03).	Transaction reference number, updated stock, rental lock.
Rental return	Reference number, returned quantity, item condition.	Restores stock (RULE-04, RULE-08), assesses late or damage fees (RULE-09, RULE-10).	Updated stock and tank status, fee statement.

Process	**Inputs**	**System Processing**	**Outputs**
Physical count	Counted quantity, item, reason code, admin approval.	Compares count with system balance (RULE-06).	Adjustment entry in the movement log.
Void or cancellation	Reference number, reason, admin approval.	Reverses the stock effects (RULE-07).	Voided transaction and audit entry.
Inventory monitoring	Stock levels after each transaction.	Checks reorder levels (RULE-05).	Low-stock alerts and overdue rental flags.
Reporting	Cutoff period and report type.	Compiles transaction and stock data.	CSV/Excel reports for sales, rentals, low stock, and tanks.

4.6 Results Interpretation and Analysis

4.6.1 Achievement of Objectives

Table 16 shows what the study produced for each specific objective in Chapter I and where it appears in this paper.

Table 16. Study Outputs per Objective

Specific Objective	Study Output	Section
1. Analyze the current process	SIPOC analysis, current-state process, and 4P root-cause diagram.	3.1.2 to 3.1.5
2. Identify process gaps and data risks	Table of problems, root causes, and operational impact.	3.1.4
3. Perform stakeholder analysis	Stakeholder analysis matrix.	3.1.1
4. Design the future-state workflow	Future-state process and before-and-after comparison.	3.2.1, 3.2.2
5. Specify requirements	Functional requirements FR01 to FR10 and non-functional requirements.	3.1.6

Specific Objective	Study Output	Section
6. Formulate business rules	Automation rules RULE-01 to RULE-10.	3.2.4
7. Construct a prototype and dashboard	Three admin screens and dashboard wireframe, based on the capstone system.	3.2.6, 3.3.1
8. Develop a test plan	Functional test cases and results (Table 18).	4.7.1
9. Formulate risk, roadmap, cost-benefit, and change plans	Risk matrix, implementation phases and schedule, cost-benefit analysis, and ADKAR change plan.	3.4

4.6.2 Expected Process Improvements

Table 17 compares the current manual process with the targets the system is designed to meet. The manual figures are estimates from the field interviews and from the assumptions in Chapter III. They have not been timed. The targets come from the requirements in Chapter III.

Table 17. Current Process Estimates and Design Targets

Measure	**Current manual process (estimate)**	**Design target**	**Basis in Chapter III**
Time to record one transaction	About 10 minutes	About 2 minutes or less	Usability requirement; cost-benefit assumption.
Monthly report preparation	About 3 hours	Under 5 minutes	Before-and-after comparison (Table 6).
Stock update after a transaction	End of day	Under 2 seconds	Performance requirement.
Rental availability	Not shown in the stock record	Rented items locked in the ledger	RULE-03.
Tracing stock discrepancies	Not possible	Every stock movement logged with user, time, and reference	FR08.

4.6.3 Financial Interpretation

The cost-benefit analysis in Section 3.4.4 gives the following preliminary figures. They depend on assumptions: 5,200 transactions per year, 10 minutes per manual transaction against about 2 minutes with the system, and a labor rate of ₱80 per hour.

- Estimated first-year cost (including 10% contingency): ₱125,278
- Estimated direct labor savings from the time saved: ₱55,466 per year
- Projected annual savings used for the ROI, which add other estimated savings (fewer discrepancies, rental conflicts, and report time): ₱133,968
- Net first-year benefit: ₱8,690, an ROI of about 6.94% and a payback period of about 11.2 months

Only the ₱55,466 labor savings come from a calculation. The remaining ₱78,502 is an estimate that still has to be supported with CareMed's records. Counting labor savings alone, the payback period would be about 27 months. The financial results should therefore be treated as preliminary and confirmed with actual figures during the pilot.

4.7 System Evaluation Results

4.7.1 Functional Test Results

Table 18 lists the functional test cases and the result of each.

Table 18. Functional Test Cases and Results

Test Case	Scenario	Expected Result	Status
TC-01	Sale quantity exceeds available stock (RULE-01)	Transaction blocked with an insufficient-stock message	Pass
TC-02	Confirm a sale (RULE-02)	Stock deducted and movement-log entry written	Pass
TC-03	Confirm a rental (RULE-03)	Item locked as Rented; deposit matches the distance rule	Pass
TC-04	Two staff try to rent the same item at once	Only one rental succeeds	Pass
TC-05	Return a rented tank (RULE-04)	Status set to Empty-For-Refill	Pass
TC-06	Stock falls to the reorder threshold (RULE-05)	Low-stock alert and reorder report entry	Pass

Test Case	Scenario	Expected Result	Status
TC-07	Physical count differs from the record (RULE-06)	Adjustment requires reason code and admin approval	Pass
TC-08	Void a transaction (RULE-07)	Stock effects reversed; reason in audit log	Pass
TC-09	Partial return (RULE-08)	Only returned quantity restored; balance outstanding	Pass
TC-10	Overdue return (RULE-09)	Late fee equals days overdue times the daily rate	Pass
TC-11	Damaged return (RULE-10)	Item set to Damaged; damage fee assessed	Pass
TC-12	Export a cutoff report (FR10)	CSV/Excel file matches the ledger	Pass
TC-13	Search history by reference, customer, date, product (FR09)	Correct records returned	Pass

4.7.2 Evaluation Against Design Targets

The functional tests in Table 18 show that the rules and transactions behave as designed. The other measures are compared with the targets in Table 17: the time per

transaction, the time needed to prepare the monthly report, the number of stock discrepancies found at the physical count, and the staff's feedback on ease of use. They come from the two-week parallel pilot described in Chapter III, in which the owner, counter staff, and inventory custodian use ATRIMS-CareMed alongside the logbook. Measured values from the pilot are not reported in this paper, so the figures in Table 17 stay as design targets and estimates until they are recorded.

4.7.3 Limitations of the Findings

All 13 functional test cases in Table 18 were run against the system and passed. These results show that the rules and transactions behave as designed. They are not measurements from live use at CareMed. The manual process times are estimates, the financial figures are preliminary, and measured pilot values are not yet reported. The expected improvements will be confirmed only after those measurements are recorded.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter summarizes the study, presents its conclusions, and gives recommendations. The study developed a Business Process Automation Plan for the Automated Transaction Recording and Inventory Management System for CareMed Medical Supply Enterprise (ATRIMS-CareMed) as a BAM-285 project. Its purpose was to address the late recording of transactions, the stock discrepancies, and the manual reporting that result from CareMed's logbook-and-worksheet process.

5.2 Summary of the Proposed Study Design and Implementation

The study followed an 11-stage Business Analysis Project Framework. Its main parts were:

1. Analysis of the current process: A stakeholder analysis, SIPOC analysis, current-state process, and 4P (People, Process, Platform, Paperwork) fishbone diagram were used to find the problems and their root causes.
2. Design of the solution: Functional and non-functional requirements, a future-state process, four system modules, ten automation rules, and a three-tier architecture were specified. The system uses Python and Flask, Bootstrap 5, and MySQL through SQLAlchemy, packaged with Docker. Three

admin-side screens were prepared based on the capstone system.

3. Planning: A test plan, risk matrix, implementation phases and schedule, cost-benefit analysis, and ADKAR-based change management plan were prepared. The 13 functional test cases were run and passed; the pilot measurements and the rollout are still to be completed.

5.3 Summary of Findings

1. Problems in the current process: CareMed's manual process results in late or omitted transaction records, no real-time view of stock, stock differences that cannot be traced, rental items promised twice, unreliable oxygen tank counts, and monthly reports that take hours. The main root cause is the deferred, memory-based way of recording, not staff carelessness.

2. What the design provides: Single-screen recording of sales and rentals with automatic totals and deposits, a real-time ledger with rental locking and tank statuses, ten automation rules, a movement log of every stock change, and exportable reports.

3. Expected improvements: Recording time of about 2 minutes or less per transaction against an estimated 10 minutes, monthly reports in under 5 minutes against about 3 hours, and immediate stock updates instead of end-of-day updates.

4. Financial estimates (preliminary): An estimated first-year cost of ₱125,278 and projected annual savings of ₱133,968, giving a net first-year benefit of ₱8,690, an ROI of about 6.94%, and a payback period of about 11.2 months. Only ₱55,466 of the savings comes from the computed labor time saved. With labor savings alone, the payback period would be about 27 months.

5. Functional test results: All 13 functional test cases (Table 18) passed, covering stock blocking, sales deduction, rental locking and deposits, tank statuses, low-stock alerts, count adjustments, voids, partial and late or damaged returns, report export, and history search.

5.4 Conclusion

Based on the analysis and design work in this study, the following conclusions are drawn:

1. The main cause of CareMed's recording and inventory problems is its reliance on deferred, memory-based recording with no real-time validation or audit trail.

2. ATRIMS-CareMed is designed to address each identified problem through point-of-transaction recording, a real-time inventory ledger, rental locking, automation rules, and a movement log that makes stock changes traceable.

3. A web-based system using Flask and MySQL on low-cost cloud hosting is technically feasible for a small

enterprise such as CareMed, since it is based on a working capstone system.

4. The estimated costs and savings suggest that the system could pay for itself in about a year if the assumed savings are achieved. This is a preliminary result that depends on validating the assumptions.

5. The nine specific objectives were addressed through the outputs listed in Table 16. All 13 functional test cases passed (Table 18). How well the system performs in daily use at CareMed still has to be confirmed.

5.5 Recommendation

The following recommendations are offered for CareMed's management, for validating the results, and for future researchers:

1. Recommendations for CareMed Management and Operations

- Use the two-week parallel pilot to confirm the measured results before retiring the logbook.
- Require every sale, rental checkout, and tank refill to be recorded in ATRIMS-CareMed at the moment it happens.
- Keep the prepaid 5G pocket Wi-Fi unit active at the counter so stock checks continue during internet outages.
- Perform monthly physical counts and investigate any difference using the movement log.

- Check each month that the automatic daily database backups can be restored.

2. Recommendations for Validating the Results

- Measure the actual time per transaction and per monthly report during the pilot to replace the estimates in this paper.
- Support the additional savings in the cost-benefit analysis with CareMed's records of discrepancies, rental conflicts, and reporting time.

3. Recommendations for Future Development

- Barcode/QR Code Integration: Add handheld scanners to speed up item selection at checkout.
- Customer Self-Service Portal: Allow customers to reserve rentals and upload payment proof.
- Native Mobile Application: Build an iOS/Android app with offline caching for delivery personnel.
- Accounting Software Integration: Export transaction data to accounting software such as QuickBooks or Xero.

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